AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I006*

Team Name: *Syntax Error*

Domain: *5, Food Security, Agriculture & Rural Transformation*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

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Problem Statement

Due to lack of data, Small Scale Farmers blindly grow crops at same time which leads to price crashes, market over supply, crop wastage , reduces cost-effectiveness and poor crop planning.

Problem Understanding

- The Problem is majorly faced by Small Scale Farmers, Retailers and Consumers.
- The Problem is caused by poor crop planning, limited access to crop data and no vision about the market demand.
- It causes market over supply, price crashes, reduces farmers profit, limiting access to consumers for fresh crops.

Existing Gaps & Challenges

- Limited integration of yield predictions, area-wise crop survey to guide farmers with crop planning.
- Lack of local crop monitoring, forecasting, data diagnostics causing high price risks and over-supply.
- Absence of affordable platform combining farm data, market insights and direct sales.

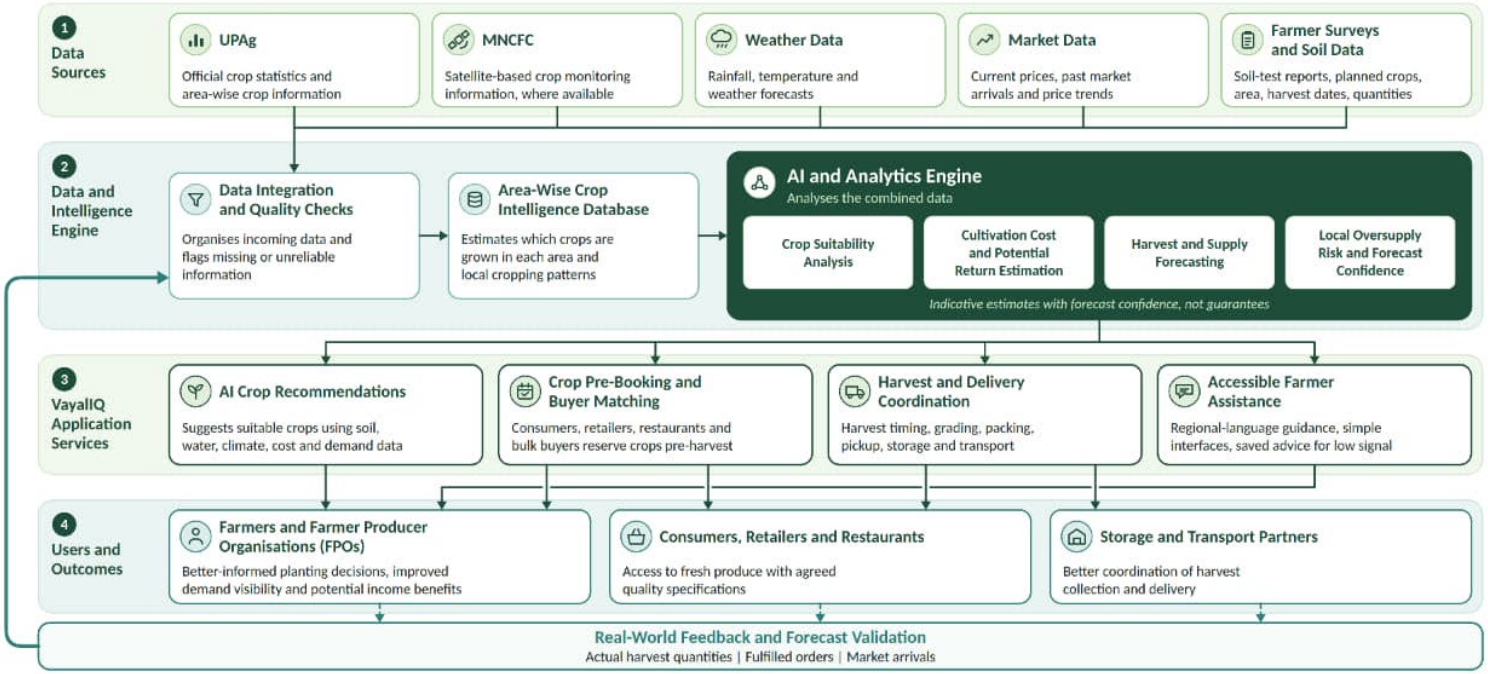
Our Proposed Solution

- VayalIQ is an affordable, AI platform that helps Small Scale Farmers in decision making, crop selection and cultivation. It integrates environmental conditions, market trends to provide them with better farming decisions.

- Our AI Support System gathers information from surveys(UPAg, MNCFC), satellite based data, analyzing weather condition and fulfilling agricultural needs and cropping patters of different regions. This platform suggests suitable crops by analyzing various conditions such as local soil conditions, water availability, cultivational expenses and potential profits. It also helps to allow consumers, retailers to book their expected crops in advance. This further helps farmers understand buyers requirements and plan their production according to market demand. The special trait of our platform is that it connects farmers directly with buyers and helps to coordinate throughout the process of production

- Our Project aims to help farmers minimize crop wastage and reducing the dependency on middle-men. Providing agricultural and market-insights.

Technical Approach



Credits : AI Generated

Tech Stack

Frontend: React.js

Database: PostgreSQL

Backend: Python (FastAPI)

AI/ML, APIs, or Hardware: Scikit-learn, forecasting models, data APIs.

Innovation & Uniqueness

- Combines location specific crop recommendations, yield prediction, crop profit analysis, market-risk and farm to consumer sales in single platform.
- It is suitable because it provides with farmer feedback, weather forecast to help Small Scale Farmers choose wise decisions on crop management.
- It connects every stage of farming from selecting the right crop to estimate production and identify the potential buyers. This approach helps farmers to understand market demand reduce over supply and create better selling opportunities and transparent transactions.

Feasibility & Scalability

- **Feasibility:** VayalIQ can be implemented using existing agricultural data, farmer service reducing the need for costly hardware at the initial phase.
- **Scalability:** The platform can expand progressively to cover additional districts, crops and farming communities by collaborating with buyers analysing the region specific data.
- **Risks:** Inaccurate data, unpredictable market risks, majorly affects the platform effectiveness. These challenges are now minimized through data verification, user-friendly regional language interface and parternship with FPOs.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: A small scale farmer uses VayalIQ to understand which crops are suitable for their area based on the historical data,soil conditions , weather patterns, cultivation costs and market demand . The platform estimates potential yield and highlights possible possible oversupply risks before planting and helping the farmer to look into various circumstances instead of relying on one past experience .

- Use case 2: A consumer, retailer or restaurant can look for anticipated harvests and pre-book the crop by selecting the desired quantity and preferred date for delivery. These pre-bookings assist farmers in assessing the demand beforehand and plan their cultivation and harvesting, accordingly, while providing the buyers improved access to fresh crops.

- Use case 3: After Harvest , Farmer producer Organisation (FPO) connect via buyers though VayalIQ and coordinate throughout quality grading , packing ,collection transport.By avoiding the unnescassary middlemen and enabling group dedliveries,the AI support system aims to improve price transparency and help small scale farmers earn better .

Impacts and Benefits

- It helps Small Scale Farmers to choose suitable crops and connect them in advance with buyers to ensure fair prices and increase the income through direct sales.
- It improves farmers livelihood further providing consumers with fresh crops and reduce crop wastage
- The measures forecast accuracy, order fulfillment, farmers total income and post harvest losses through a pilot study

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.3	Zero Hunger	Supports Small scale farmers by improving productivity, market access and potential earnings.
SDG 12	Target 12.3	Responsible Consumption and Production	Helps reduce food wastage through demand-based crop planning, advance booking and timely delivery

Individual Member Contribution

Member	Contribution or responsibility
Akhilesh Aravind J	Problem identification, stakeholder research and existing agricultural challenges
Sharvesh J B	Solution design and crop-planning workflow
Gaurav T	Technical architecture, technology selection, feasibility, cost estimation and use cases
Roahith P	Impact analysis, SDG mapping

Yukti ID

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